

Machine Learning Capstone Project Steps

Objective

Build an end-to-end machine learning project within **two hours** using real-world data. Participants may freely use AI tools such as **ChatGPT, Claude, Gemini, and GitHub Copilot**, but they must understand and explain their final solution.

Project Steps

Step 1: Define the Problem

Choose an industry-relevant problem involving:

- Classification
- Regression
- Clustering
- Forecasting
- Recommendation
- Anomaly detection

Clearly mention the business problem, target users, expected output, and practical use case.

Step 2: Collect the Data

Use a real dataset with at least **10,000 rows**.

Data may come from:

- APIs
- Web scraping
- Open-source platforms
- Government portals
- Kaggle
- Industry or organisational data

Mention the dataset source and collection method.

Step 3: Clean and Prepare the Data

Perform the required preprocessing:

- Remove duplicates
- Handle missing values
- Correct data types
- Detect and treat outliers
- Encode categorical variables
- Scale or normalise numerical features
- Handle class imbalance
- Remove irrelevant or leakage-prone columns

Step 4: Perform Exploratory Data Analysis

Use statistics and visualisations to identify:

Machine Learning Capstone Project Steps

- Data distributions
- Correlations
- Trends
- Patterns
- Outliers
- Relationships between variables
- Target-variable behaviour

Explain the key insight from each important chart.

Step 5: Engineer and Select Features

Create useful features and remove unnecessary ones.

Possible methods include:

- Derived variables
- Date and time features
- Binning
- Transformations
- Interaction features
- Feature importance
- Correlation-based selection
- Dimensionality reduction

Avoid target leakage.

Step 6: Split and Validate the Data

Use an appropriate validation strategy:

- 70:30 split
- 80:20 split
- Train-validation-test split
- K-Fold Cross-Validation
- Stratified K-Fold
- Time-series validation

Fit preprocessing steps only on the training data.

Step 7: Train Multiple Models

Test up to **10 suitable machine learning algorithms**.

Examples:

- Logistic or Linear Regression
- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Machine Learning Capstone Project Steps

- Support Vector Machine
- K-Nearest Neighbours
- Naive Bayes
- Neural Network
- K-Means or DBSCAN for clustering

Use CPU or GPU resources where appropriate.

Step 8: Tune Hyperparameters

Improve the best-performing models by tuning parameters such as:

- Learning rate
- Number of trees
- Tree depth
- Number of estimators
- Regularisation
- Batch size
- Epochs
- Dropout
- Class weights
- Number of clusters

Use Grid Search, Random Search, Optuna, or another suitable method.

Step 9: Evaluate and Compare Models

Use metrics appropriate to the task.

Classification: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix

Regression: MAE, RMSE, MSE, R-squared

Clustering: Silhouette Score, Davies-Bouldin Index

Also check:

- Overfitting
- Underfitting
- Cross-validation performance
- Training time
- Prediction time

Create a comparison table and chart for all models.

Step 10: Present Results and Create a Dashboard

Show:

- Best-performing model
- Actual versus predicted values
- Confusion matrix or cluster visualisation
- Feature importance

Machine Learning Capstone Project Steps

- Error analysis
- Business interpretation
- Model limitations
- Recommendations for improvement

Create a simple dashboard using Streamlit, Power BI, Tableau, Plotly, Dash, or Gradio.

Step 11: Publish on GitHub

Create a public GitHub repository containing:

- Project overview
- Business use case
- Dataset source
- Data-cleaning steps
- EDA visualisations
- Models tested
- Evaluation results
- Final comparison
- Dashboard or demo link
- README file
- requirements.txt
- Instructions to run the project

Do not upload passwords, API keys, confidential data, or personal information.

Final Deliverables

Submit:

1. Public GitHub repository link
2. Jupyter Notebook or Python project
3. EDA and visualisations
4. Model comparison table
5. Final model evaluation
6. Dashboard or working demo
7. README with findings, limitations, and business recommendations